

Requirements Engineering Visualization: A Systematic Literature Review

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Abstract—Requirements Engineering (RE) is a decision-centric activity which is highly data-intensive. The results of this process are known to have key impact on the results of the project. As known from the experience in other fields and disciplines, visualization can potentially provide more insights into data, information and knowledge studied. While research in the area of information visualization and its application to software engineering has rapidly increased over the last decade, there is only a limited amount of studies addressing the usage and impact of visualization techniques for RE activities. In this paper, we report on the results of a Systematic Literature Review (SLR) related to RE visualization. Extending the established SLR process by the usage of grounded theory for the encoding of papers, we synthesize 18 usage patterns. Even though there are punctual applications, there is a clear deficit on a holistic perspective across the different RE activities. As another conclusion, we derive the clear need for more research on visualization support in particular for tackling requirements uncertainty, requirements verification, and modelling, as well as Non-Functional Requirements (NFRs).

I. INTRODUCTION AND BACKGROUND

Requirements Engineering (RE) activities and tasks such as identifying projects stakeholders, exploring their needs and expectations, communicating requirements and goals, and managing and monitoring requirements changes are the most data-intensive and media-rich activities in every software development project [1]. In addition, decision-making lies at the heart of early RE activities, and most of these decisions, such as allocating requirements to a specific release or transferring business objectives to technical specifications, are often made with inevitable uncertainties about the final cost, schedule, user needs and expectations, and the systems' functionalities [2]. The high level of uncertainty in requirements decisions, coupled with the massive, heterogeneous, and dynamic volumes of information in the RE process, makes this process the most error-prone activity in every software development project [1]. However, it is believed that visualization techniques which aim to facilitate information flow in RE activities and increase awareness of project stakeholders can improve the RE process, and consequently provide practical solutions to reduce misunderstandings and communication gaps related to identifying and communicating the requirements [3].

Software Visualization (SV) is a field of Software Engineering (SE) that uses computers to visualize the software development process and its related artifacts (e.g. requirements,

design documentations, bug reports), and it can be effectively used to support human reasoning and insight, and to foster the productivity of software development processes [4]. According to the results of an empirical experiment in which 111 researchers in SE were asked about the importance of SV, Rainer Koschke [5] reported that 82% of the participants believed SV is important for the success of the software development process. Likewise, Bassil and Keller [6] conducted a study on a SV tool with 107 participants mostly from industry. Based on the results of this study, *better understanding of software, increasing productivity and quality of the development process, and complexity management* are the main advantages for which practitioners use SV.

The importance of RE visualization in effectively communicating and managing stakeholder expectations and conflicts has been extensively discussed by researchers [1], [4], [5], [7]. RE visualization, such as visualizing RE decisions and their consequences and relationships, is critical to alleviate conflicts among stakeholders and to manage requirements changes. While gathering, categorizing, and analyzing the existing visualization techniques can be of use to practitioners and researchers seeking to apply these techniques in their work, there has been no effort to systematically gather, classify, and analyze the RE visualization literature. In order to fulfill this research gap in the literature and to summarize research on the application of visualization techniques in the area of RE, we conducted a Systematic Literature Review (SLR). Moreover, conducting this SLR is in line with the research agenda of our previous work [8] (presented at RENext! 2015) to visualize the uncertainty inherent in requirements debt decisions.

This review seeks to analyze, classify, and present the common visualization techniques which have been developed to support various dimensions of RE, such as requirements activities, stakeholders and domains (Table I). Since various paradigms of visualization, such as Data Visualization (DV), Information Visualization (IV), and Knowledge Visualization (KV) differ regarding their ability to transfer and communicate data, in this review we classified studies from various visualization perspectives: (1) *Data Visualization*: graphical representation of unprocessed information [9] (e.g. points, lines or bars). (2) *Information Visualization*: Card et al. [10] defined IV as “The application of *computer supported*, visual representations of abstract data to amplify cognition”. (3)

TABLE I
VARIOUS DIMENSIONS OF RE

RE Stakeholders [12], [13] -Users: who will actually operate and interact with the system -Developers: who design, build, and maintain the system, such as analyst, designer, programmer, tester, and so on -Decision-makers: who have executive power and control over projects decisions to build the system, such as the managers of the development and users organizations, business, and product managers. -Clients: who will pay for the system
RE Domain [14] -Problem: requirements that represent the stakeholders' needs and expectations, -Solution: requirements that represent the subsequent layers (e.g. system requirements)
RE Activities [12], [15] -Elicitation: Identifying the users and project sponsors needs and expectations, projects stakeholders, sources of requirements, and understanding the application domain -Modelling: Representing the RE process as well as a whole range of RE artifacts. Enterprise, data, behavioural, and domain modelling are some general categories of RE modelling. -Communication: Circulating the identified requirements among project's stakeholders. -Validation and Verification: Evaluating the completeness and correctness of the elicited requirements and artifacts. -Evolution: Managing and monitoring requirements changes (e.g. adding, deleting, or modifying requirements)

Knowledge Visualization: Burkhard [11] defined the concept of KV as an approach that “examines the use of visual representations to improve the transfer of knowledge between at least two persons or group of persons”. The two significant contributions of this review to the RE visualization body of knowledge are:

- 1) This SLR reports on the process of identifying, gathering, classifying, synthesizing, and analyzing a set of relevant papers on RE visualization techniques based on predefined inclusion and exclusion criteria, and rigorous data extraction and analysis methods.
- 2) This SLR applied the grounded analysis technique to make a body of knowledge for RE visualization. This evidence can be of use to researchers and practitioners from various perspectives, such as identifying the research gaps in the area of RE visualization and elaborating on the existing RE visualization techniques and their relation with various visualization paradigms and various RE perspectives.

The rest of this paper is structured as follows: Section II describes the details of our review methodology, including research questions, inclusion and exclusion criteria, our search strategies for both manual and automated searches, and the methodology for data extraction and analysis. In Section III, we discuss the results and findings of the review based on our research questions. Threats to the validity of our review results are discussed in Section IV. In Section V, we present related work and discuss the scope and limitations briefly. Finally, in Section VI, we conclude the review and provide recommendations for future research.

II. REVIEW METHOD

In this review we followed the guidelines provided by Kitchenham and Charters [16] and Zhang et al. [17] for conducting a SLR and identifying relevant studies in software engineering. This section elaborates on the steps we followed during our review process:

A. Research Questions

This review aims at addressing the following Research Questions (RQs):

RQ1: What visualization techniques are used to support various dimensions of RE?

The RQ1 aims at investigating how and to what extent the various dimensions of RE (i.e. activities, stakeholders, and domains) are supported by visualization.

RQ 2: What are the different purposes of the visualization techniques applied in RE?

This question aims to identify the main functionality of the visualization techniques used to support RE (e.g. coordination, recall, attention).

RQ 3: What are the different visualization types used to support RE activities?

This question seeks to find out the visualization types (e.g. sketch, diagram, image, map, object, etc.) that are most used to visualize various aspects of RE.

RQ 4: How much evaluation is done to analyze the adaption of RE visualization techniques in practical contexts?

This question guides us to identify the maturity of the existing RE visualization techniques. The results of this question help researchers identify new topics of empirical studies, and enable practitioners to investigate the maturity of the proposed RE visualization techniques.

B. Inclusion and Exclusion Criteria

The following criteria were used as inclusion and exclusion criteria for the selection of the primary studies in our review:

1) *Exclusion Criteria:* we excluded publications that meet the following criteria: (1) publications that were not related to RE visualization (2) tutorials, proposals, and position papers and other non-peer reviewed publications (3) peer-reviewed publications that do not meet the minimum requirements of any type of visualizations: (V_1) the results of the visualization should be based on the raw data. This requirement implies that image processing and photography, in which the source of data is an image are not treated as visualization. (V_2) The outcome of the visualization should be one (or more) image(s) that represent and communicate the data. (V_3) The results of the visualization should be readable and understandable by a viewer (e.g. decision makers, requirements engineers) [18]. Visual variables such as *size*, *shape*, *orientation*, *colour*, *value*, and *texture* are basic characteristics of visual representations that must meet these requirements, as stated by Carpendale [19]. Considering these requirements, we excluded publications that only addressed UML and Goal modeling as a visualization technique (27 papers in total). While these techniques are widely being used by researchers and practitioners to model

system requirements, they consider only one of these required visual variables (*shape*), and, as stated by Diehl [4], the visual efficiency of these models is low (V_2 and V_3 are not met).

2) *Inclusion Criteria*: We included publications that: (1) addressed visualization techniques to represent various dimensions of RE (2) provided some details about their visualization technique and its application in RE (e.g. journal, conference, and workshop papers) (3) considered the visualization prerequisites discussed in Section II-B1.

C. Search Strategy

To retrieve the maximum number of relevant literature for our review, we conducted both manual and automated searches. To this end, we followed the steps provided by Zhang et al. [17] to identify the relevant studies in SE as follows:

1) *Step 1- Manual Search*: we first identified a set of publication venues and databases related to the areas of SE, RE, and information visualization. Next, these venues were evaluated and finalized by two experts from the areas of SE and information visualization independently, who determined if each of the identified venues should be included for the manual search step. These include: *RE*, *REV* (the Requirements Engineering Visualization workshop, with the main focus on visualizing and representing RE activities), *ICSE*, *IEEE VIS*, *VisSoft*, *EASE*, *FSE*, *JSS*, *IST*. The included papers from this venue were a great input for the process of frequency analysis (section II-C2) that was performed to identify the search strings required for the automated search step. Next, we manually reviewed the proceedings and journals of these venues and judged their relevance to our research questions based on their title, keywords, and abstract. In situations where the inclusion decision was not possible based only on these components, we reviewed the conclusion, headers/sub-headers, and references of the papers. At the end of this step, we identified 16 papers that addressed RE visualization.

To evaluate the reliability of our inclusion and exclusion decisions we used the *Cohen's Kappa* statistic [20], which calculates the degree of agreement between two raters, in our case, the first two authors of the paper. The calculated Kappa value was 0.87, which shows significant agreement, as stated by Landis and Koch [21]. All of the identified papers in this step are used to form the Quasi-Gold Standard (QGS), an approach for evaluating the quality of the keywords string [17].

2) *Step 2- Identifying Search Strings*: in this step, we used text mining techniques to extract the most frequently used words in the papers identified in the manual search step. To this end, we used the **Text Mining Package (tm)**¹ of **R**². As stated by Keshav [22], the title, abstract, introduction, section/subsection headings (not their content), and conclusion of a paper represent its general idea and main contributions. Thus, before conducting the *frequency analysis*, we applied the main steps of text pre-processing on these parts of the 16 papers identified in the manual search step (Section II-C1).

¹<http://tm.r-forge.r-project.org/>

²<https://www.r-project.org/>

TABLE II
SEARCH TERMS

Category 1	Category 2	Category 3
Visualization • •	Requirements Engineering •	Functional •
Visualisation • •	Requirement •	Non-functional •
Model(ling) • •	Goal •	Software Engineering • •
Design(ing) • •	NFR(s) •	
Interaction •	Release Planning •	
Communication •	Stakeholder •	
Diagram • •		
Collaboration •		

For the purpose of pre-processing, we conducted the following steps iteratively: (1) *Manual Transformation* (e.g., removing hyphens which appeared after converting the pdf files to plain text). (2) *Removing numbers and punctuations* (e.g. years and in-text reference numbers). (3) *Removing stop words* (e.g. articles, conjunctions, and common verbs). (4) *Stemming* (e.g. reducing *visualization*, *visualizing*, and *visualize* to their root **visual**).

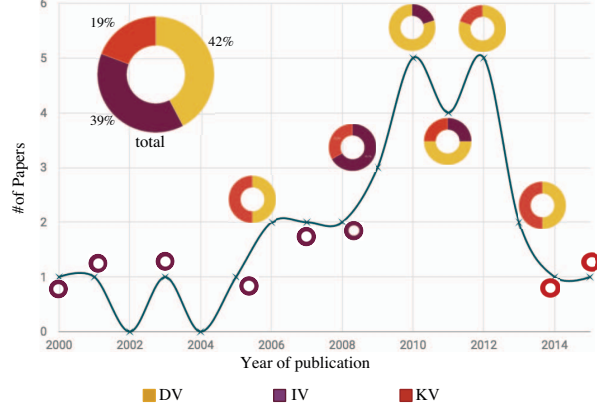
Next, we performed the frequency analysis on the resulting text files and extracted the most frequent words of the 16 identified papers. Table II lists these words and demonstrates how the search terms can be combined to form the search strings using Boolean operations. Each search string can be formed by using a Boolean “AND” operator between search terms of different columns with the same colour, and a Boolean “OR” operator between different search strings. For instance, “*Visualization AND Requirements Engineering*” **OR** “*Communication AND Stakeholder AND Software engineering*” are two valid search strings for the automated search step.

3) *Step 3- Automated Search*: in this step, we first queried *Google Scholar* (GS) using the search terms listed in Table II. However, Dean et al. [23] recently performed an experiment to evaluate if GS is sufficient to be used for systematic reviews, and concluded that *GS does not meet the required search standards for systematic reviews, and is not able to identify all known and relevant papers in a specific area*. Thus, at the next step we searched the following electronic databases to find more relevant papers: ACM Digital Library, ScienceDirect, SpringerLink, and IEEE Xplore. This yielded 559 articles published within the last 16 years (2000- 2016), the studies before 2000 were only focused on UML modeling as a technique of RE visualization.

4) *Step 4- Quasi-Gold Standard Evaluation*: To evaluate the reliability and quality of the search terms we identified during the manual search step, we followed the guidelines provided by Zhang and Babar [24] and formed the Quasi-Gold Standard for years 2006-2010 (five years of REV workshop), which includes 16 papers. After performing the automated search, we retrieved 15 of these studies. Thus, the *quasi-sensitivity* is 94%, which represents an acceptable performance for our search process.

5) *Step 5- Snowballing*: To complement the search results obtained, we used snowballing by scanning the reference list (backward snowballing [25]) of identified papers. Furthermore, there were papers that have been published after the selected

Fig. 1. Number of studies per type of visualization they supported per year.



papers, so we performed citation tracking (forward snowballing [25]) to identify relevant papers citing the papers in the previous step. At the end of this stage, we added three more papers to the list of the primary studies of this review, a total of 26 papers for our analysis.

D. Data Extraction and Synthesis Method

During this step, we used the following two methods to extract and synthesize data from each of the 26 primary studies included in this review:

(1) *Grounded Analysis*: During this step, we applied a revised version of the grounded theory [26] methodology for conducting a rigorous literature review and followed the steps we implemented in our previous SLR [27] to extract data from each of the 26 primary papers included in this review. We analyzed all of the included papers line by line to extract concepts, which involves high volumes of qualitative data. To help with this process, we used Saturate³, which is a web-based open coding tool that enables traceability between codes and data. Due to limited space, we only report the results of the open coding and axial coding steps (Table III).

(2) *Data Extraction Form*: To organize the process of data extraction, we created three data collection forms, in which the following data points should be captured by the reviewer of each paper: (1) visualization paradigm (e.g. data, information, or knowledge visualization) (2) various dimensions of RE (i.e. RE activities, stakeholders, and domain) (3) various perspectives of the visualization that have been addressed for RE (i.e. function, audience, goal, and type). More details about these parameters are listed in Tables IV, V, and VI.

III. RESULTS AND FINDINGS

A. Overview of Papers

This section gives an overview on the results of analyzing the primary studies included in this SLR in terms of their published year and visualization paradigm. As illustrated in Figure 1, more than 50% of these studies are published in the period 2010-2012 and there is an increasing trend on supporting knowledge visualization in RE from 2006.

³www.saturateapp.com

1) *Visualization Paradigm*: Since RE is a high communication and decision intensive activity, knowledge visualization can help to improve the communication among stakeholders and reduce the communication gaps and conflicts among end-users and technical stakeholders of software development projects. However, based on the results of this SLR, only 19% [P22-26] of the studies addressed this visualization paradigm, while 39% ([P12-21]) and 42% ([P1-11]) of the studies addressed information and data visualization, respectively.

TABLE III
OVERVIEW OF OPEN CODES EXTRACTED FROM THE PAPERS INCLUDED IN THIS REVIEW. NUMBERS IN PARENTHESES SHOW THE NUMBER OF OCCURRENCES OF EACH CODE IN THE INCLUDED PAPERS

Categories	Open codes
Requirements Communication (42)	Distributed Requirements Engineering (10) [P1, P9, 10], (# of papers: 3)
	Stakeholders Communication (14), [P1-3, P9-10, P12, P21-22, P25] (# of papers: 9)
	Communication Gap (18) [P2, P6-7, P9, P12-13, P17, P20, P23, P25-26] (# of papers: 11)
Requirements Evolution (47)	Requirements Change (23) [P6, P11-12, P14, P21, P24-26], (# of papers: 8)
	Tracing Requirements (11) [P10, P14-15, P18, P23, P26], (# of papers: 6)
	Requirements Relationships (13) [P6, P10-11, P18, P23, P25-26], (# of papers: 7)
NFRs (19)	Non- Functional Requirements (19) [P4-5, P13, P15, P18, P20, P22, P25] (# of papers: 8)
Requirements Inspection (27)	Requirements Uncertainty (2) [P23-24], (# of papers: 2)
	Requirements Validation (8) [P6, P10, P17, P26], (# of papers: 4)
	Requirements Verification(3) [P12, P24], (# of papers: 2)
Requirements Planning (27)	Requirements Prioritization (15) [P4, P7, P11, P13, P17, P22-25], (# of papers: 9)
	Requirements Planning (12) [P1, P11, P22-24], (# of papers: 5)

With respect to Knowledge Visualization (KV), S. Feather et al. [P1] provided two visualization types (i.e. diagram(s) and Map) to help project stakeholders to understand the *how* and *why* of their decisions. For instance, they used a 2D scatter plot to represent the likelihood and impact of a requirement's risk(s). They also used Kiviat chart to represent the comparison among several risk mitigation decisions.

2) *Requirements for RE Visualization*: we used grounded analysis to extract data from the primary of this review. By using this technique, we coded all of these papers to explore the main areas (dimensions) of RE in terms of their need of visualization support. After similarity analysis of these codes (open codes and axial codes), we found (Table III): *Requirements evolution* and *communication* were among the most RE activities with visualization support, with each of them being 30% and 26% of the extracted codes assigned to each category, respectively. *Requirements inspection* and *planning* come next, with 19% of the codes for each category. Visualizing *Non-Functional Requirements (NFRs)* had less support, with only 12%. More details about the results of applying grounded analysis are available on the website of

the first author of the paper ⁴.

Finding 1: Applying grounded analysis revealed that Requirements Planning and Inspection and NFRs have the strongest deficit of visualization support.

B. RQ1- Visualization Support for Various Aspects of RE

In this section, we report on the results of our review regarding the current status of visualization support for various aspects of RE. Table IV lists the studies that supported various dimensions of RE, as well as some examples of visualization techniques proposed to support these dimensions.

Based on the results of our review, as demonstrated in Figure 2 and Table V, representing a project's requirements and their related artifacts (*what*) is addressed in all of the included papers in our SLR. Studies that addressed the process (*how*) of performing RE activities in their proposed visualization techniques come next with 19 (73%) studies. Representing the motivation of performing a specific RE task or making a requirement decision, as well as representing the roles (*who*) which are involved in performing the task or making the decision are addressed in 13 (50%) and 8 (31%) studies, respectively. Software development teams and companies are often geographically far from their users and customers. While visual representation of RE tasks and decisions can alleviate the difficulties of communication and coordination in globally distributed RE teams, visualization support for distributed RE is only addressed in 4 (15%) of the studies.

1) *RE Activities*: As listed and illustrated in Table IV, *requirements elicitation* is the most supported task by visualization among all RE activities, with 16 (62%) of the papers. Visualization support for other RE activities: *requirements verification*, *modelling*, *evolution*, and *communication* are addressed in 10 (38%), 8 (31%), 8 (31%), and 5 (19%) of the papers, respectively. Figure 2 depicts visualization support for various combinations of RE activities.

We performed further analysis on the results of this RQ and extracted the common **RE Visualization Patterns** in the form of $\langle \text{content}, \text{focus} \rangle$, where *content* shows each of the RE activities, and *focus* denotes the *What*, *How*, *Why*, *Where*, and *Who* components of each visualization technique. The details of these visualization patterns are illustrated in Figure 2. As illustrated in this figure, none of the visualization techniques proposed by the included studies in our review addressed all of the RE activities. Further, two combinations of (*elicitation, modelling*) and (*elicitation, communication, validation*) are the most frequent patterns addressed by these studies.

Finding 2: Based on the results of the analysis we performed on the results of RQ1, we extracted the common RE visualization patterns in the form of $\langle \text{content}, \text{focus} \rangle$, where **content** shows each of the RE activities, and **focus** denotes the *What*,

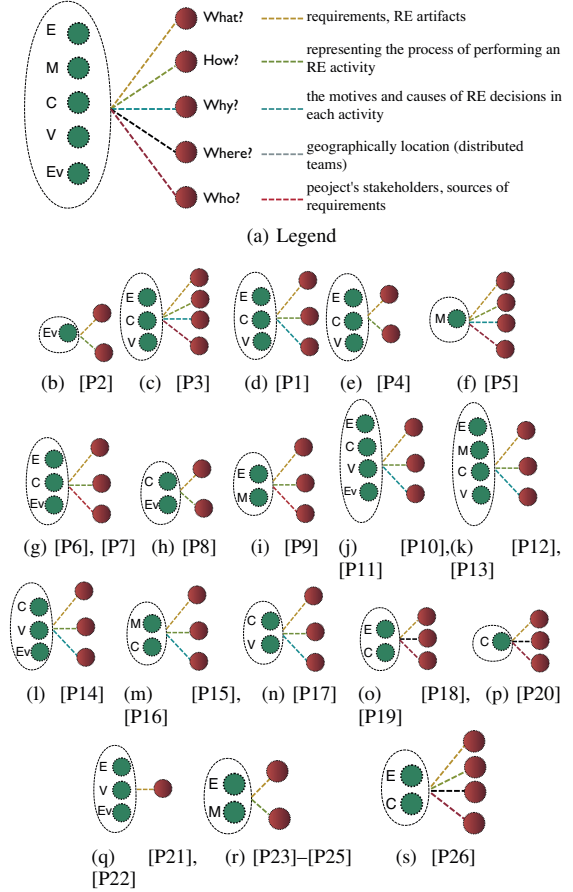


Fig. 2. The common RE visualization patterns ($\langle \text{content}, \text{focus} \rangle$) [E: Elicitation, M: Modeling, C: Communication, V: Validation, Ev: Evolution]

How, Why, Where, and Who components of each visualization technique.

This finding can be used as a reference for selecting visualization techniques for a specific combination of RE activities and visualization goal (focus).

2) *Requirements Engineering Stakeholders*: The results of our review show that “developers” and “decision makers” are the most supported stakeholder group in the studies included in our review (18 (69%) studies for each category). Among all the studies, 15 (58%) studies targeted end-users in their visualization approach, and *customers* have less visualization support, with only 8 (31%) of the studies.

3) *Requirements Engineering Domain*: As illustrated in Table IV, while 88% of the studies supported the problem domain in their visualization techniques, only 50% of the studies addressed the solution domain.

C. RQ2- Different Visualization Functions Applied in RE

To address this RQ, we used *data extraction forms* to explore different visualization purposes (functions) proposed by researchers and practitioners. We used the following func-

⁴<http://wcm.ucalgary.ca/zshakeri/home>

TABLE IV
RQ1- VISUALIZATION SUPPORT FOR VARIOUS DIMENSIONS OF RE

	Dimensions /Related papers	The Proposed Visualization Techniques
RE Activities	Elicitation (E) [P1-2, P4-6, P8, P10, P12-13, P16-20, P23-24]	• [P19]: A Stacked Pareto chart and horizontal bar charts to visualize the requirements priority and the correlation of each stakeholders votes with the resulting priorities, respectively.
	Modelling (M) [P3, P5-7, P10-11, P13, P20]	• [P15]: (1) The application of visual objects (glyphs) and graphical variables (i.e. colour, thickness) to model product-line requirements relationships and inter-dependencies. (2) Highlighting the decisions with the greatest number of inter-dependencies to visualize the strength of the dependencies on the decisions as well as the difficulty of decisions.
	Communication (C) [P9, P11, P19, P22, P26]	• [P26]: A visual interactive representation of requirements to involve a large number of geographically distributed users in the process of RE. This approach improves communicating the requirements in two ways: (1) filtering requirements based on their geo-coordinate, and (2) filtering requirements based on the user-assigned keywords, which represents the similarities and dependencies between requirements.
	Verification (V) [P6, P12-15, P17-18, P22-23, P26]	• [P12]: A visualization technique to represent NFR patterns including problem patterns (e.g. threats, and vulnerable and undesirable situations) and objective patterns (e.g. different interpretations of NFRs that are provided by a project's stakeholders and represents their conflicts in defining these NFRs).
	Evolving (Ev) [P2, P8, P12, P14, P18, P21, P24, P26]	• [P10]: An interactive visualization technique, which enables a project's stakeholders to directly manipulate requirements in real time during their decision making process.
Stakeholders	End users (Eu) [P3, P6-7, P9-10, P12-13, P15, P17, P19-22, P25-26]	• [P21]: A visualization technique for end-users with different levels of familiarity with computer applications, by which users can select the features of the system from a list of predefined visual objects and make an image of the product based on their needs and preferences. • [P4]: A 3D visualization of requirements to reduce the gap between developers and end users and to validate the requirements in the earlier stages of the development process.
	Developers (D) [P1-2, P4-6, P8-9, P13-14, P17-23, P25-26]	• [P7]: Integrating visual objects, specific to scientific software development, into the requirements models to reduce the learning effort for developers who build scientific applications to successfully create their requirements model and have concrete ideas of the system before starting the development tasks.
	Decision-makers (Dm) [P1, P3-7, P9, P11, P13, P15, P19-26]	• [P6]: A three dimensional, space-filling, and growing pyramid, which provides various views of the selected requirements for each group of projects stakeholders. This visualization technique mainly represents the following parameters for each of the system requirements: $(R = \{S_n, W, Ref\})$, while S_n , W , and Ref denote the stakeholders to which requirement belongs, requirements scope in terms of its attainment level, and the level of the requirements refinement, respectively.
	Customers (C) [P3, P7, P9, P12, P16, P21, P25-26]	• [P16]: A separate graphical object for the system's customers to differentiate them from end-users and business stakeholders and to involve them in RE activities and consequently make their needs and operations more clear and understandable (Figure 3(e)).
Domain	Problem (P) [P1, P3-7, P9-13, P15-26]	• [P17]: A Visual Requirements Analytics (VRA) technique for improving risk assessment decisions by using the following graphical variables on cohesive bar and arc graphs: (1) <i>colour</i> for representing the level of requirements attainment, (2) <i>height</i> for representing the correlation of a requirement category with other requirements categories in terms of risk components, (3) <i>width</i> for representing the ratio of related risks to a requirements, and (4) <i>order</i> for prioritizing requirements based on their impact on the risk components.
	Solution (S) [P2, P4-6, P8-9, P13-14, P18-19, P21, P23, P25]	• [P23]: A visualization technique to model NFRs, which impacts various requirements decisions, regardless of if they are system-level or software level. • [P11]: A visual tool (DREAMER) to represent and model the traceability and coverage of both requirements and design options during the development process.

tions, proposed by Burkhard [11] to classify the visualization functions proposed by included studies in our review.

- **Coordination:** coordinating and managing individuals during the communication process.
- **Recall:** improving the understandability and remembrance ability of viewers by using conceptual diagrams.
- **Maps:** following cartographic standards to represent hierarchy of data (e.g. a ground layer represents the project's context and individual elements on this layer represent the project's milestones, risks, or resources).
- **Motivation:** inspiring viewers to take actions.
- **Attention:** grabbing viewer attention by representing trends, outliers, and in general all the characteristics that impact viewer decisions.

- **Elaboration:** providing more clarification about visual representations.
- **New Insights:** creating new insights by showing relationships between elements and by demonstrating the context patterns and details.

Table V lists these functions, the studies that addressed each function, and some examples of visualization techniques proposed for each function. Among all the studies, 15 (58%) studies addressed the attention function, which is the most addressed function. Elaboration and Motivation come next, with 13 (50%) studies each. Coordination and insight are addressed in only 10 (38%), and 9 (35%) studies, respectively. Recall has the least support with only 4 (15%) studies.

Moreover, we analyzed the proposed RE visualization techniques from their *audience* perspectives, such as *individuals*,

TABLE V
RQ2- DIFFERENT PURPOSES OF VISUALIZATION TECHNIQUES APPLIED IN RE

Visualisation Functions	Some Samples of the Proposed Visualization Techniques
Coordination [P9, P11, P19, P22, P26]	• [P10]: An interactive Visual Requirements Analytics (VRA) approach which helps stakeholders to overview the system requirements, detect inconsistencies and anomalies, and relate heterogeneous artifacts and concerns. • [P20]: A notation-based visualization technique to represent the flow of requirements by visualizing both informal and formal communications, and to coordinate the communication among stakeholders in global RE teams.
Attention [P2, P4-7, P11, P13-15, P18, P20, P22-24, P26]	• [P23]: The application of visual variables (e.g. size, colour, texture, and value) to represent four quality attributes such as <i>trustability</i>, <i>performance</i>, <i>feasibility</i>, and <i>certainty</i>, respectively.
Recall [P3, P5, P8, P10]	• [P5]: A set of semantical transparent modelling symbols, which allow end-users without a goal modelling background to participate in RE activities. • [P8]: Feature Survival Charts (FSC) as to represent dynamic scope changes of projects and past project scoping activities.
Motivation [P1-2, P6, P11, P14-15, P20-24, P26]	• [P2], [P3]: highlighting the flow of project resources and requirements prioritization options to help project stakeholders to compare multiple alternative lists of requirements together, and select the next release based on the current status of resources and to alert teams about requirements changing priorities.
Elaboration [P6, P8, P10, P15, P17-23, P25-25]	• [P13]: Highlighting goals and other elements within a conflicting path, and sources of these conflict to help decision-makers to easily understand conflicts and their intention, and to identify the domain trade-offs. • [P11]: A bifocal view of diagrams, which allows viewers to focus on a particular object (e.g. requirement, design object, or artifact) to explore its connection with other objects of the visualization.
Insights [P1, P13-15, P22-26]	• [P14]: The application of visual variables such as colour and value to propose a traceability visualization technique that analyzes and represents the candidate links between project requirements, a visual feedback to validate these links, and represents requirements with high architectural significance and high level of design risk.

group, organization, and network. Based on the results of this analysis, all the included studies in this SLR targeted individuals in their visualization techniques (e.g. requirements analysts, architects, developers, or end-users). Supporting RE activities in group (or RE teams) comes next with 19 (73%) studies [P3-7, P9, P11, P13-16, P19-26]. Visualization support at the organization or network level are addressed by only 7 (27%) [P7, P9, P16, P19-20, P22, P25] and 4 (15%) [P1, P9, P16, P19] studies, respectively.

Finding 3: Following the results of “grounded analysis”, requirements communication and change require more visualization support. However, coordination function, which support this requirement, has been addressed by only 19% of studies. Thus, there is a clear need for more visualization techniques that support this function.

D. RQ3- Different Visualization Types Used to Support RE Activities

To address this RQ, we used *data extraction forms* to explore visualization types proposed to support various dimensions of RE (See Table I for more details about these dimensions). Table VI gives an overview of these visualization types, the list of studies addressed each type, and some visualization techniques proposed for each type. Overall, looking at Table VI, 81% (21) of the included studies in this SLR used diagrams (e.g. Sankey, Kiviat and bar charts, and 2D scatter plot) to represent the structural relationships among various activities and artifacts of RE. Among all the studies, 11 (42%) proposed their own graphical *objects* to add more visual intuitiveness to the visual representations of RE activities and artifacts. *Map* and *interactive* representation of RE activities come next, with

7 (27%) studies for each type. *Image* and *Sketch* were used in only 4 (15%) and 2 (8%) studies, respectively.

With respect to *storytelling* as a visualization type that represents a sequence of events, decisions, or changes, each of which can contain various types of visualization (e.g. image, diagram, or sketch), text or video or any combination of these components [28]. Based on the results of our analysis, none of the included studies (0%) in this review addressed the application of this technique in RE.

Finding 4: Visualization of Requirements change is not well supported by literature

Visualization types that can manage various complexities and challenges inherent in the process of requirements change, are not well supported by researchers and practitioners. For example, by using storytelling visualization type, various aspects of requirements change can be represented by narrations explaining the current state and the changes as they occur over time. Moreover, *filling gaps* and *continuity* in the storytelling approach, as stated by Gershon and Page [28], increases the audience’s awareness of the transition between various states of the system as well as the sequence of requirements decisions.

E. RQ4- Evaluation of Proposed RE Visualization Techniques

To answer this research question, we used the following hierarchy to evaluate the evidence level [29] of the proposed visualization techniques: **(E1)** no evidence, evidence obtained from: **(E2)** demonstration, **(E3)** expert opinion, **(E4)** academic studies, **(E5)** industrial studies, **(E6)** industrial practice. Based on the results of our analysis, the main strategies for empirical evaluation are *E3* (27%) [P1, P6, P12, P15-16, P20, P22], *E5* (27%) [P4-5, P7-8, P10, P17-18], *E6* (19%) [P13, P23-26], *E4* (15%) [P2-3, P9, P21], *E1* (8%) [P11, P19], and

TABLE VI
RQ3- DIFFERENT VISUALIZATION TYPES USED TO SUPPORT RE ACTIVITIES

Visualization Types	
Image and Sketch [P2-3, P12, P17]	[P4]: The application of 3D and animation visualization techniques to develop a 3D image of the final product and to validate its features with the end users. This approach reduces the total development effort by validating requirements in the earlier stages of the project.
Diagram [P1, P4-11, P13-16, P18-20, P22-26]	[P3]: Combination of Sankey diagrams and Parallel Coordinates in a multiple tree layout to represent the information flow of requirements prioritization and release planning processes. [P1] The application of bar, Kiviati, 2D scatter plot, and charts and range to convey the risk status of project requirements and candidate decisions to mitigate these risks and to represent the consequence of uncertainties in the input data about requirements risks and their mitigation decisions.
Map [P2, P19, P21-23]	[P1]: TreeMap visualization to show requirements and sub-requirements hierarchy, relative importance, and attainment status of each requirement by using <i>size</i> and <i>colour</i> parameters respectively. [P22]: Emotional Density Map (EDM) to demonstrate the intended emotional requirements in designing and developing video games by using grayscale shading and graphical objects.
Object [P1-3, P7-9, P11-13, P18, P26]	[P18], [P22]: The application of graphical objects to manage communicating requirements in distributed teams (Fig 3(a, b)) and to capture and represent the primary and secondary emotional requirements. [P5]: Proposing a set of semantical transparent symbols for <i>Goal-oriented modeling</i> by conducting an exploratory study with 104 participants (without any background knowledge of goal modeling). Figure 3 (c,d). [P16]: Using <i>semiotic clarity</i> (i.e. defining one notational element for each of the concepts in requirements modeling) to differentiate business stakeholders from end-users, and to represent the concepts of goal, soft-goal, threats, and requirements risk mitigation (Figure 3 (e)).
Interactive Visualization [P6, P10, P12, P17-18, P22, P26]	[P3]: An interactive brushing technique to help project decision makers to interactively reveal the relationship in data (e.g. tapping on each of the elicited requirements highlights all of the releases it belongs to, as well as the representing stakeholder vote about that specific requirement).
Storytelling –	

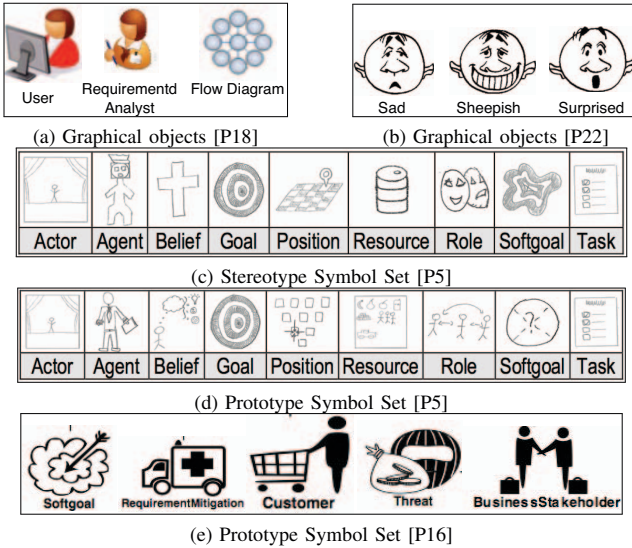


Fig. 3. Application of graphical objects (Glyph) in RE visualization

E2 (4%) [P14]. Given these data and according to the above classification, most of the studies used preliminary evaluation methods to evaluate their proposed visualization techniques. No evidence occurred in two studies [P11, P19]. Additionally, almost all of these studies employed exactly one type of evaluation, which implies that there is a clear need for more evidence to evaluate the quality of the existing RE visualization techniques.

IV. THREATS TO VALIDITY

In this SLR, to provide a comprehensive review on the studies on RE visualization, we followed the guidelines presented by Kitchenham [16] and Zhang et al. [17] for performing a SLR in software engineering. However, there are threats to the validity of the results and findings of our review.

Firstly, the completeness of the list of the primary studies included in this SLR highly depends on the keywords and the capability limitations of the employed digital libraries and search engines. While we used an objective search terms elicitation approach (i.e. text mining and frequency analysis) to reduce the risks of subjective search terms, the search standards and syntax vary among the employed search engines and libraries in this review, thus we may have missed some studies related to the research questions of our review.

Secondly, the robustness and comprehensiveness of the framework we used to classify the studies may affect the results and findings of this review. We used the framework proposed by Burkhard [11] to classify the studies based on their visualization function, goals, types, and audience. To avoid a framework with insufficient paradigms for the purpose of classification, we selected a framework related to knowledge visualization, which covers the other visualization approaches (i.e. data and information visualization). Moreover, to reduce the classification bias, all of the classification results were checked by the first two authors.

V. RELATED WORK

With respect to the related literature reviews on the area of visualization support for RE, so far, to the best of our knowledge, there is only one study provided by Amyot and

Mussbacher [30] aimed at describing and analyzing the first ten years of the User Requirements Notation (URN). The authors did a study inspired by the systematic literature review approach, in which 281 research papers related to URN were analyzed and synthesized to provide a historical overview of the development of URN together with research implications and application domains for this notation. However, the review is limited to URN and does not address other aspects of RE, such as RE activities and stakeholders. To our knowledge, the SLR we provided in this paper is the first SLR that aims to analyze and synthesize the existing evidence on RE visualization, and to provide insight into the existing research trend in this area as well as future research implications.

Regarding other types of reviews, such as survey or literature review, there are different reviews existing that addressed RE visualization. Cooper Jr et al. [3] conducted a survey of the research papers that were presented from 2006 to 2008 at the RE Visualization (REV) workshop to represent the evolution of the research trends in the area of RE over a specific time period. According to the results of this survey, many opportunities exist to develop visualization for the early steps of the RE phase (e.g. understanding the context and undertaking the groundwork necessary for the RE process) as well as verification and validation tasks, which is in line with one of the main findings of this SLR. Gotel et al. [1], [7] provided two reviews on the primary areas in which visualization techniques and artifacts can be applied to support RE activities. The results of this review highlight the need for visualization techniques to support the multi-dimensional nature of requirements and to support various diagnostic activities and decision making tasks during software development. However, these reviews have been conducted about 7 to 9 years ago. Thus, new studies are required to analyze and to synthesize the existing research works in this area and to define the research trends and implications for other researchers.

VI. CONCLUSION AND RESEARCH IMPLICATIONS

Visualization techniques are increasingly being used by researchers and practitioners to understand and to manage RE decisions and activities. By conducting this SLR, we gathered, classified, and analyzed these techniques from various perspectives to address the key research questions mentioned in Section II-A. To this end, we initially identified 559 studies by querying the employed search engines and digital libraries (Table VII). After applying the inclusion and exclusion criteria that were identified at the beginning of our review process (Section II-B) we selected 26 studies as the primary studies of our review, while the others were not related to the research questions. The findings of this review tell us the following: (1) More investigation and research are needed to support knowledge visualization in the area of RE, (2) There is no visualization support for the whole of the RE lifecycle, (3) Among all RE activities, requirements communication and evolution (Table IV) have less visualization support, (4) Visualizing NFRs and requirements uncertainties need more

TABLE VII
AUTOMATED SEARCH RESULTS

Library/publisher	#Retrieved	#QGS	#Selected
IEEE Xplore	311	13	16
ACM DigitalLibrary	28	0	2
SpringerLink	46	1	5
ScienceDirect	31	1	0
Others	143	0	3
Total	559	15	26

investigation, (5) There is a clear need for more effort in addressing the following visualization types in the context of RE: *storytelling*, *maps*, and *interactive visualization*, (6) There is a clear need for more visualization support for distributed RE, (7) The results of RQ4 show that further evaluation of the existing visualization methods is mandatory to provide better evidence regarding the quality and maturity of proposed visualization methods.

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